

	RIPHAH INTERNATIONAL UNIVERSITY, LAHORE CAMPUS. SCHOOL OF COMPUTING & INNOVATION
	FYP Scope and Feature Implementation

Group Member Names & SAP IDs		Project title	
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This form is intended to finalize the project features for each of groups.

FYP FEATURE DETAILS

Application Type	Feature/Sub-feature listing	Requirements
Web/ Mobile	1. User Portals (CRUD Operations) - Admin Dashboard: Manage students, teachers, and violations. - Discipline Incharge Dashboard: Review and confirm detected or reported violations. - Student App: View profile, fines, and report violations. 2. Role & Rights Management - Different access levels for Admin, Incharge, and Students. 3. Multi-Criteria Search - Search by student, violation type, date, or department. 4. Notifications & Reports - Real-time alerts, daily summaries, and violation statistics.	- Web (MERN) and Mobile (Flutter) applications. - CRUD for all users and violations. - Role-based authentication using Supabase/JWT. - MongoDB for records, Supabase for storage & notifications.
AI/ Research based	1. Facial Recognition - Identifies students using trained face models. 2. Weapon Detection - Detects firearms or knives using YOLO/TensorFlow. 3. Fight Detection - Recognizes aggressive actions via video frame analysis. 4. Auto Violation Logging & Fines - System records violator details and assigns fines automatically.	- Uses TensorFlow, OpenCV, YOLOv8 for detection. - Integrated with Node/Express backend. - Research based on Computer Vision & Deep Learning. - Enhances campus safety through real-time AI monitoring

Sr. No	Module	User Feature	Sub-feature	Requirement	Description
1	Authentication	Students, Admin, Discipline Incharge	Login	The system shall validate user credentials and provide secure login access.	Secure authentication for all users.
		Students, Admin, Discipline Incharge	Forgot Password	The system shall allow password reset through email verification.	Enables account recovery.
		Students, Admin, Discipline Incharge	Sign Up	The system shall register users after successful email verification.	Prevents fake and duplicate accounts.
		Students, Admin, Discipline Incharge	JWT/Firebase Authentication	The system shall provide role-based authentication and session management.	Ensures secure access control.
2	Student Management	Admin	Student Registration	The system shall register students and store academic details.	Creates student records.
		Admin	Facial Image Capture	The system shall capture multiple images of students.	Dataset generation for face recognition.
		Admin	Model Training	The system shall train and update facial recognition models.	Improves identification accuracy.
		Admin	Profile Management	The system shall create, update, and delete student profiles.	Maintains student information.
3	Violation Detection	System, Admin	Facial Recognition	The system shall identify students from live camera feeds.	Matches faces with registered database.
		System, Security	Weapon Detection	The system shall detect firearms or knives using AI models.	Generates immediate alerts.
		System, Security	Fight Detection	The system shall identify aggressive behavior and fights.	Enhances campus safety.
		System	Dress Code Detection	The system shall detect uniform violations through computer vision.	Identifies dress code breaches.
		System	Violation Logging	The system shall automatically store violation details.	Saves evidence, time, and student information.

4	Violation Reporting	Students	Manual Reporting	The system shall allow students to report violations with evidence.	Encourages student participation.
		Students	Evidence Upload	The system shall allow image/video submission.	Provides proof of incidents.
		Students	Cool-down Timer	The system shall restrict report submissions within a specific interval.	Prevents spam reporting.
		System	Report Validation	The system shall verify report authenticity using AI and metadata.	Filters false reports.
		System, Admin	Reward Assignment	The system shall assign reward points for valid reports.	Encourages responsible reporting.
5	Fine & Reward System	System, Admin	Auto Fine Generation	The system shall automatically assign fines based on violation severity.	Enforces discipline policies.
		Admin	Fine Management	The system shall review, update, or waive fines.	Allows administrative control.
		Students	Reward Points	The system shall maintain student reward records.	Motivates positive behavior.
6	Dashboards	Admin	Admin Dashboard	The system shall display all users, violations, and analytics.	Centralized management panel.
		Admin	User Management	The system shall perform CRUD operations on users.	Manages students and teachers.
		Discipline Incharge	Violation Review	The system shall allow approval or rejection of violations.	Human verification of incidents.
		Discipline Incharge	Reports Dashboard	The system shall provide violation summaries.	Weekly and monthly reports.
		Students	Student Dashboard	The system shall display personal profile, fines, and rewards.	Student self-service portal.
7	Notifications & Alerts	Admin, Discipline Incharge	Real-Time Alerts	The system shall send notifications for critical violations.	Immediate awareness of incidents.
		Admin, Discipline Incharge	Summary Notifications	The system shall provide daily and weekly reports.	Helps monitor campus activity.
		Students	Fine Notifications	The system shall notify students regarding fines and reports.	Keeps students informed.

8	Analytics & Reports	Admin	Violation Analytics	The system shall provide charts and statistics for violations.	Identifies trends and repeat offenders.
		Admin	Report Generation	The system shall generate weekly and monthly reports.	Supports decision making.
		Admin	Export Reports	The system shall export reports in PDF and Excel format.	Documentation and record keeping.
9	Security & Privacy	System	Data Encryption	The system shall encrypt facial and personal information.	Protects sensitive data.
		All Users	Role-Based Access Control	The system shall restrict access according to user roles.	Prevents unauthorized access.
		System, Admin	Audit Logs	The system shall record all user activities.	Ensures accountability and traceability.
10	Future Enhancements	Admin	Predictive Analysis	The system shall predict potential violations using behavioral patterns.	Preventive AI monitoring.
		Admin	ERP Integration	The system shall integrate with university ERP systems.	Centralized academic management.
		Admin	Parent Notifications	The system shall notify parents about serious violations.	Improves communication.
		Admin, Security	Voice Alerts	The system shall trigger emergency voice announcements.	Real-time emergency response.

ADMIN FUNCTIONALITIES

Sr. No	Module / Feature	Functionality	Detailed Description
1	Authentication	Login, Sign Up, Forgot Password	Admin can securely log in, create accounts, and reset passwords through email verification and role-based authentication.
2	Student Management	Student Registration	Register new students by entering academic and personal information.
		Facial Image Capture	Capture multiple facial images of students for AI model training.
		Model Training	Train and update facial recognition models using captured images.
		Profile Management	Create, update, view, and delete student records and associated facial data.
3	Violation Management	View Violations	Access all AI-detected and manually reported violations.
		Edit Violations	Modify violation details, severity level, or evidence information.
		Delete Violations	Remove invalid or duplicate violation records from the system.
4	Violation Detection	Dress Code Monitoring	Monitor violations related to improper uniforms or dress code breaches.
		Weapon Detection Monitoring	View and verify weapon detection alerts generated by AI models.
		Fight Detection Monitoring	Monitor physical fights and aggressive behavior detected by the system.
		Identity Verification	Verify detected individuals using facial recognition results.
5	Fine & Reward System	Fine Review	Review fines automatically assigned to students.
		Fine Modification	Increase, decrease, or waive fines based on circumstances.
		Reward Approval	Approve reward points assigned to students for valid reports.
6	Dashboard	Administrative Dashboard	View complete system overview including users, violations, alerts, and statistics.
		Incident Monitoring	Monitor real-time violation activities across campus.
7	Notifications & Alerts	Critical Alerts	Receive immediate notifications for weapon detection, fights, and major incidents.
		Summary Notifications	Receive daily, weekly, and monthly activity summaries.
8	Analytics & Reports	Violation Analytics	View violation trends by department, category, and student.
		Report Generation	Generate detailed violation and disciplinary reports.
		Export Reports	Export reports in PDF and Excel formats for documentation.
9	User Management	Manage Students	Add, edit, activate, deactivate, or remove student accounts.
		Manage Teachers	Add, edit, activate, deactivate, or remove teacher accounts.
		Manage Roles	Assign and modify user roles and permissions.
10	Security & Privacy	Role-Based Access Control	Manage user permissions and access rights.
		Audit Logs	View complete logs of user activities and system actions.
		Data Security	Ensure secure storage and handling of personal and facial data.
11	Future Enhancements	Predictive Analysis	View AI predictions regarding potential repeat offenders.
		Parent Notifications	Configure alerts sent to parents for serious violations.
		Voice Alert Management	Manage emergency voice announcements and alerts.

DISCIPLINE INCHARGE / TEACHER FUNCTIONALITIES

Sr. No	Module / Feature	Functionality	Detailed Description
1	Authentication	Login & Forgot Password	Secure login and password recovery using email verification.
2	Violation Review	AI Violation Review	Review violations detected through facial recognition and AI monitoring.
		Student Report Review	Examine violations reported manually by students.
3	Approve / Reject Violations	Incident Validation	Verify evidence and determine whether violations are valid.
		Approval Process	Approve confirmed violations for fine generation.
		Rejection Process	Reject false or invalid violation reports.
4	Reports	Weekly Reports	Generate weekly summaries of campus violations.
		Monthly Reports	Generate monthly disciplinary reports.
		Department Reports	View violation statistics for specific departments.
5	Notifications	New Incident Alerts	Receive alerts whenever a new violation is detected.
		Critical Alerts	Receive urgent notifications regarding fights or weapon detection.
6	Analytics	Student Behavior Analysis	View summarized statistics of student disciplinary records.
		Violation Trends	Analyze common violation categories and trends.
7	Security & Access	Role-Based Permissions	Access only authorized modules and student information.
		Secure Data Access	Ensure confidentiality of disciplinary records.
8	Future Enhancements	Parent Collaboration	Coordinate with admin regarding parent notifications.
		Voice Alert Support	Participate in emergency voice alert procedures.

STUDENT FUNCTIONALITIES

Sr. No	Module / Feature	Functionality	Detailed Description
1	Authentication	Login	Secure login using registered credentials.
		Sign Up	Register a new account with email verification.
		Forgot Password	Reset password using email verification.
2	Profile Management	View Profile	View personal information and academic details.
		Violation History	Access personal violation records and evidence.
3	Manual Violation Reporting	Report Submission	Capture images/videos and submit violation reports.
		Evidence Upload	Upload supporting evidence for reported incidents.
4	Cool-down Timer	Report Restriction	Prevent repeated submissions within a predefined time interval.
5	Reward Points	Reward Tracking	View earned reward points for valid reports.
		Reward History	Track reward point transactions and achievements.
6	Fine Management	Fine Viewing	View fines assigned for verified violations.
		Fine History	Access complete history of penalties and payments.
7	Student Dashboard	Personal Dashboard	View profile, violations, rewards, and fines in one place.
		Activity Summary	Monitor reporting and disciplinary activity.
8	Notifications	Fine Alerts	Receive notifications when fines are assigned or modified.
		Report Updates	Receive status updates regarding submitted reports.
		System Notifications	Receive important announcements and alerts.

9	Security	Personal Data Access	Access only personal records and information.
		Role-Based Security	Restrict unauthorized access to other users' data.

SYSTEM / AI FUNCTIONALITIES

Sr. No	Module / Feature	Functionality	Detailed Description
1	Facial Recognition	Student Identification	Identify registered students using trained facial recognition models.
		Face Matching	Compare detected faces against the student database.
2	Weapon Detection	Object Detection	Detect firearms, knives, and dangerous objects using YOLOv8.
		Alert Generation	Trigger alerts when weapons are detected.
3	Fight Detection	Aggression Detection	Detect physical altercations using action recognition models.
		Event Classification	Classify suspicious aggressive activities.
4	Auto Violation Logging	Violation Recording	Automatically record violation type, student ID, timestamp, and evidence.
		Evidence Storage	Store captured images and videos securely.
5	Fine Generation	Automatic Fine Assignment	Calculate and assign fines based on violation type and severity.
6	Report Validation	AI Verification	Validate authenticity of student-submitted reports.
		Metadata Analysis	Analyze timestamps, images, and location information for verification.
7	Notifications	Real-Time Alerts	Send notifications to admins and discipline incharges.
		Emergency Alerts	Generate immediate alerts for critical incidents.
8	Model Training	Continuous Learning	Retrain facial recognition models when new students are registered.
		Model Updates	Improve AI accuracy through periodic updates.
9	Data Encryption	Secure Storage	Encrypt facial images, student information, and violation records.
		Privacy Protection	Ensure compliance with data security requirements.
10	Predictive Analysis	Behavioral Prediction	Analyze violation history to identify potential repeat offenders.
		Risk Assessment	Generate risk scores for preventive disciplinary actions.